Department of Electronics and Communication Engineering
THE LNM INSTITUTE OF INFORMATION TECHNOLOGY, JAIPUR

(DEEMED TO BE UNIVERSITY)
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Microwave Engineering Theory
2nd Year ECE Project
Date of Submission: 29/04/23

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Name of Project 1 (Part-A):

I. Question 7:

Design a stepped-impedance low-pass filter having a cutoff frequency of 3 GHz, and a fifth-order 0.5 dB equal-ripple response. Assume R_0=50Ω, Z_l = 15Ω, and Z_h=120Ω.

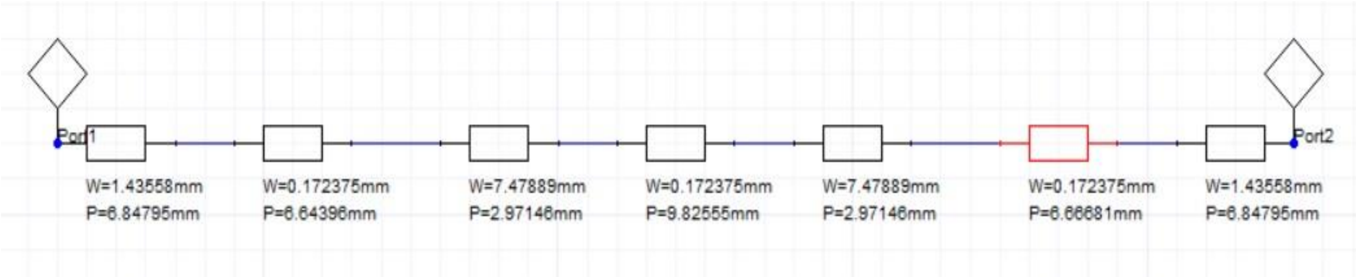
- Find the required electrical lengths of the five sections
- Use Ansoft Designer to plot the insertion loss from 0 to 6 GHz.
- Lay out the microstrip implementation of the filter on an FR4 substrate having $\varepsilon_r = 4.2$, d = 0.079 cm, tanδ = 0.02, with copper conductors 0.5 mil thick. Use ADS to plot the insertion loss versus frequency in the passband of the filter.

II. Simulation Platform:

ANSOFT DESIGNER SOFTWARE

III. Design and Analysis:

(a) Screenshots of schematics



(b) calculations

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Microwave Engineering Assignment

Part A [Question 7]

$f_c = 3GHz$

$N = 5$

$T_{ypo} = 0.5dB$, Equal ripple response

$R_o = 50\Omega$

$Z_l = 15\Omega$

$Z_h = 120\Omega$

Electrical length $= \beta l = \frac{Z_0 L}{Z_h}$ for inductor $\beta l = \frac{Z_l C}{Z_0}$ for capacitor

Prototype Elements

$g_0 = 1$ $g_1 = \frac{a_1}{F_2}$ $g_k = \frac{a_{k-1}\, a_k}{b_{k-1}\, g_{k-1}}$ $g_{n+1} = \begin{cases} 1 & \text{for } n \text{ odd} \\ \coth^2(F_1) & \text{for } n \text{ even} \end{cases}$

where $F_1 = \frac{1}{4} \ln\left\{\coth\left(\frac{L_r}{17.372}\right)\right\}$

$F_2 = \sinh\left(\frac{2F_1}{N}\right)$

$a_k = 2\sin\left\{\frac{(2k-1)\pi}{2n}\right\}$ $k = 1, 2, -- n$

$b_k = F_2^{\,2} + \sin^2\left(\frac{k\pi}{n}\right)$ $k = 1, 2, -- n$

$F_1 = \frac{1}{4} \ln\left\{\coth\left(\frac{L_r = 0.5}{17.372}\right)\right\} = 0.887$

$F_2 = \sinh\left(\frac{2 \times 0.887}{5}\right) = 0.362$

$a_1 = 2\sin\frac{\pi}{10} = 0.617$ $a_3 = 2\sin\left(\frac{5\pi}{10}\right) = 1.44$

$a_2 = 2\sin\left(\frac{3\pi}{10}\right) = 1.617$ $a_4 = 2\sin\left(\frac{7\pi}{10}\right) = 1.614$

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$$a_5 = 2\sin\left(\frac{9\pi}{10}\right) = 0.62$$

$$b_1 = (0.362)^2 + \sin^2\left(\frac{\pi}{5}\right) = 0.131 + 0.345 \Rightarrow 0.476$$

$$b_2 = (0.362)^2 + \sin^2\left(\frac{2\pi}{5}\right) = 0.131 + 0.904 = 1.035$$

$$b_3 = (0.362)^2 + \sin^2\left(\frac{3\pi}{5}\right) = 0.131 + 0.905 = 1.036$$

$$b_4 = (0.362)^2 + \sin^2\left(\frac{4\pi}{5}\right) = 0.131 + 0.346 = 0.477$$

$$b_5 = (0.362)^2 + \sin^2\left(\frac{5\pi}{5}\right) = 0.131 + 0 = 0.131$$

$$g_1 = \frac{a_1}{F_2} = \frac{0.617}{0.362} = 1.704 \qquad g_2 = \frac{a_1 a_2}{b_1 g_1} \Rightarrow \frac{0.617 \times 1.617}{0.476 \times 1.704} \Rightarrow 1.23$$

$$g_3 = \frac{a_2 a_3}{b_2 g_2} \Rightarrow \frac{1.617 \times 1.99}{1.035 \times 1.23} = 2.52 \qquad g_4 = \frac{a_3 a_4}{b_3 g_3} \Rightarrow \frac{1.99 \times 1.619}{1.036 \times 2.52} \Rightarrow 1.23$$

$$g_5 = \frac{a_4 a_5}{b_4 g_4} \Rightarrow \frac{1.619 \times 0.62}{0.477 \times 1.23} \Rightarrow 1.71$$

① $\beta l = \frac{Z_0 L_1}{Z_h} \Rightarrow \frac{50 \times 1.704}{120} \Rightarrow 40.70^\circ$

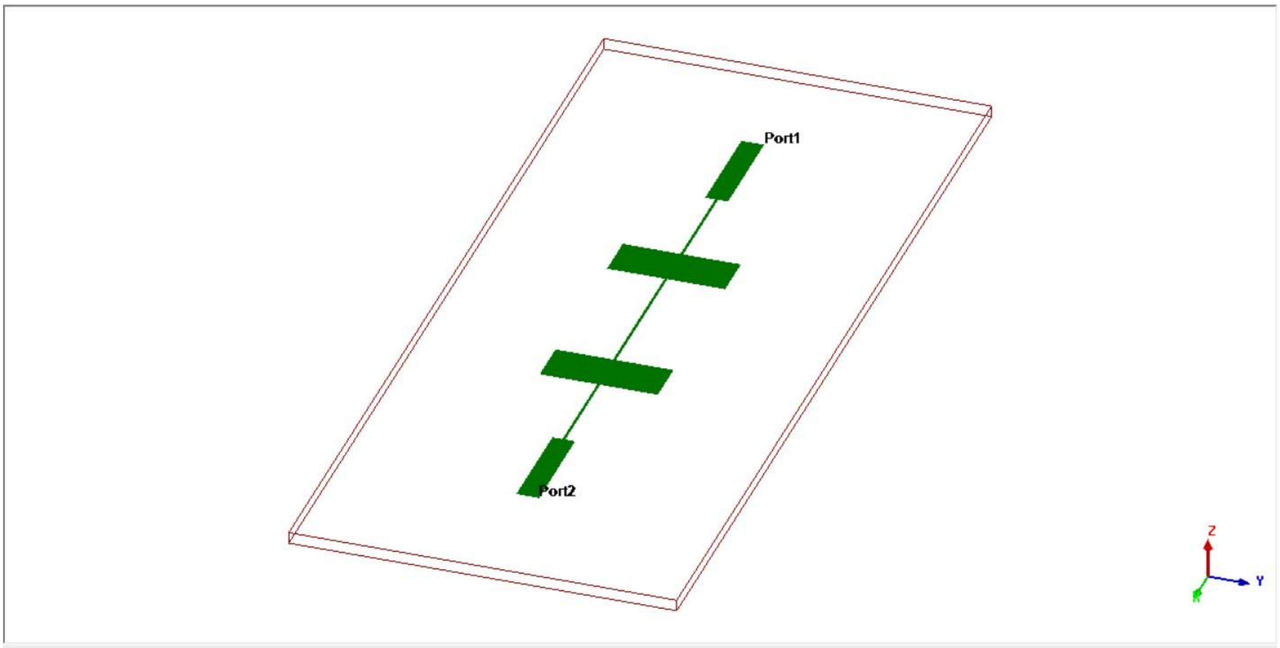
② $\beta l = \frac{Z_L C_2}{Z_h} \Rightarrow \frac{15 \times 1.23}{50} = 0.369 \Rightarrow 21.15^\circ$

③ $\beta l = \frac{Z_0 L_3}{Z_h} \Rightarrow \frac{50 \times 2.52}{120} \Rightarrow 1.05 \Rightarrow 60.19^\circ$

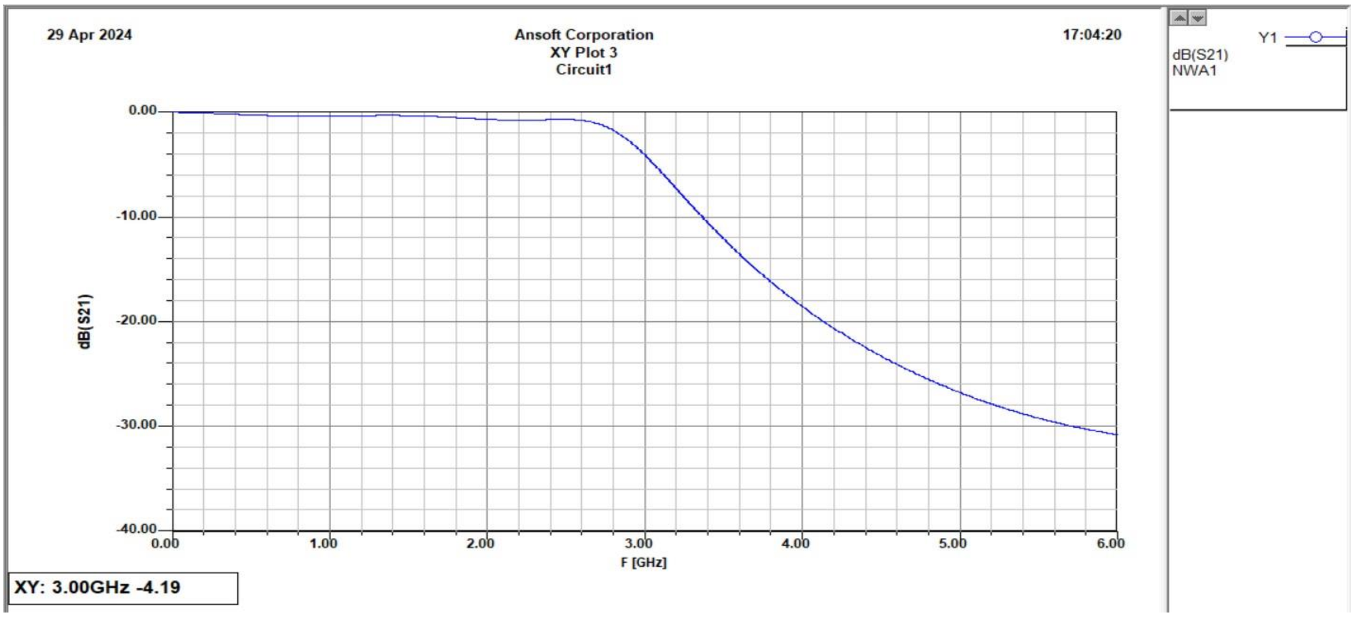
④ $\beta l = \frac{Z_l C_4}{Z_0} \Rightarrow \frac{15 \times 1.23}{50} \Rightarrow 0.369 \Rightarrow 21.15^\circ$

⑤ $\beta l = \frac{Z_0 L_5}{Z_h} \Rightarrow \frac{50 \times 1.71}{120} \Rightarrow 0.7125 \Rightarrow 40.84^\circ$

(c) 3D Layout



(d) **Analysis of the results.**



IV. Conclusion

We successfully implemented 0.5db equal ripple filter with cutoff frequency of 3GHz.

Name of Project 2 (Part-B):

I. Question 11:

Design a Wilkinson power divider with a power division ratio of $P_3/P_2=1/5$, and a source impedance of 50Ω.

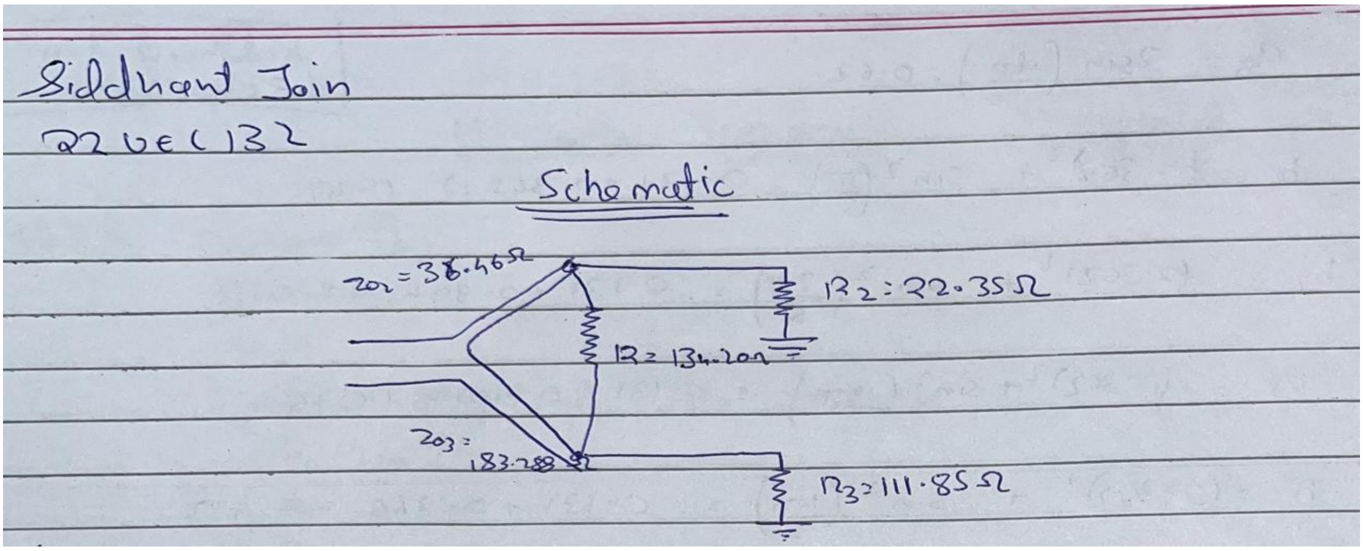
- Show all calculations
- Design the schematic diagram.
- Use Ansoft Designer to plot the S_{12}, S_{21}, S_{32} versus frequency.

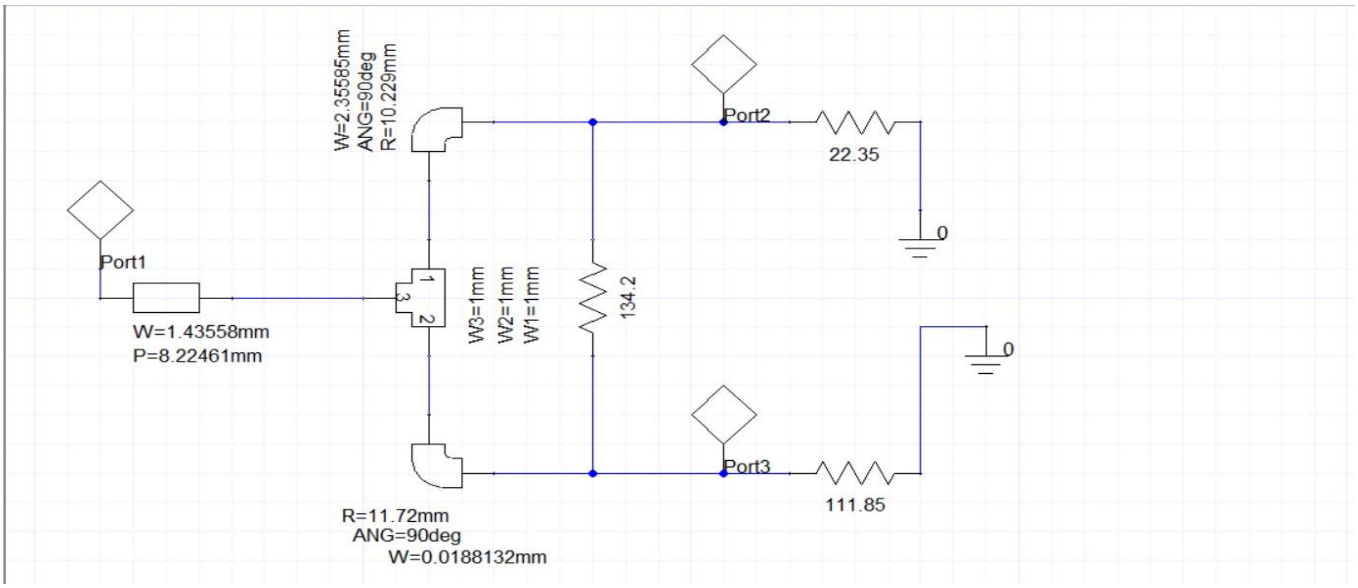
II. Simulation Platform:

ANSOFT DESIGNER SOFTWARE

III. Design and Analysis:

(a) Screenshots of all schematics





(b) Screenshots of results.

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Part B [Question 11]

$\frac{P_3}{P_2} = \frac{1}{5}$ $\quad k^2 = \frac{1}{5}$, $\quad k = \frac{1}{\sqrt{5}} = 0.447$ $\quad Z_0 = 50\Omega$

$Z_{o_2} = Z_0 \sqrt{k(1+k^2)}$

$\Rightarrow 50 \sqrt{(0.447)(1.19)} \Rightarrow \boxed{36.46\ \Omega}$

$Z_{o_3} = Z_0 \sqrt{\frac{1+k^2}{k^3}} \Rightarrow 50\sqrt{\frac{1+(0.447)^2}{(0.447)^2}} \Rightarrow 50 \sqrt{\frac{1+0.199}{0.0893}} \Rightarrow 50\sqrt{13.43}$

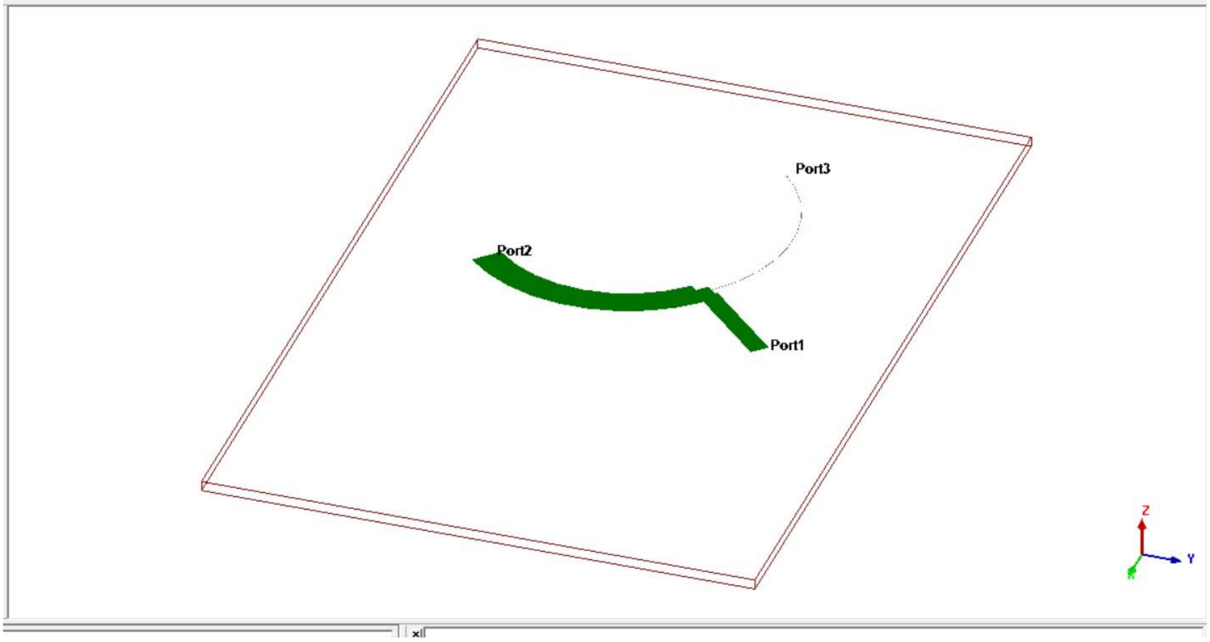
$\boxed{Z_{o_3} = 183.288\ \Omega}$

$R = Z_0\left(k + \frac{1}{k}\right) \Rightarrow 50\left[0.447 + \frac{1}{0.447}\right] \Rightarrow \boxed{134.30\ \Omega}$

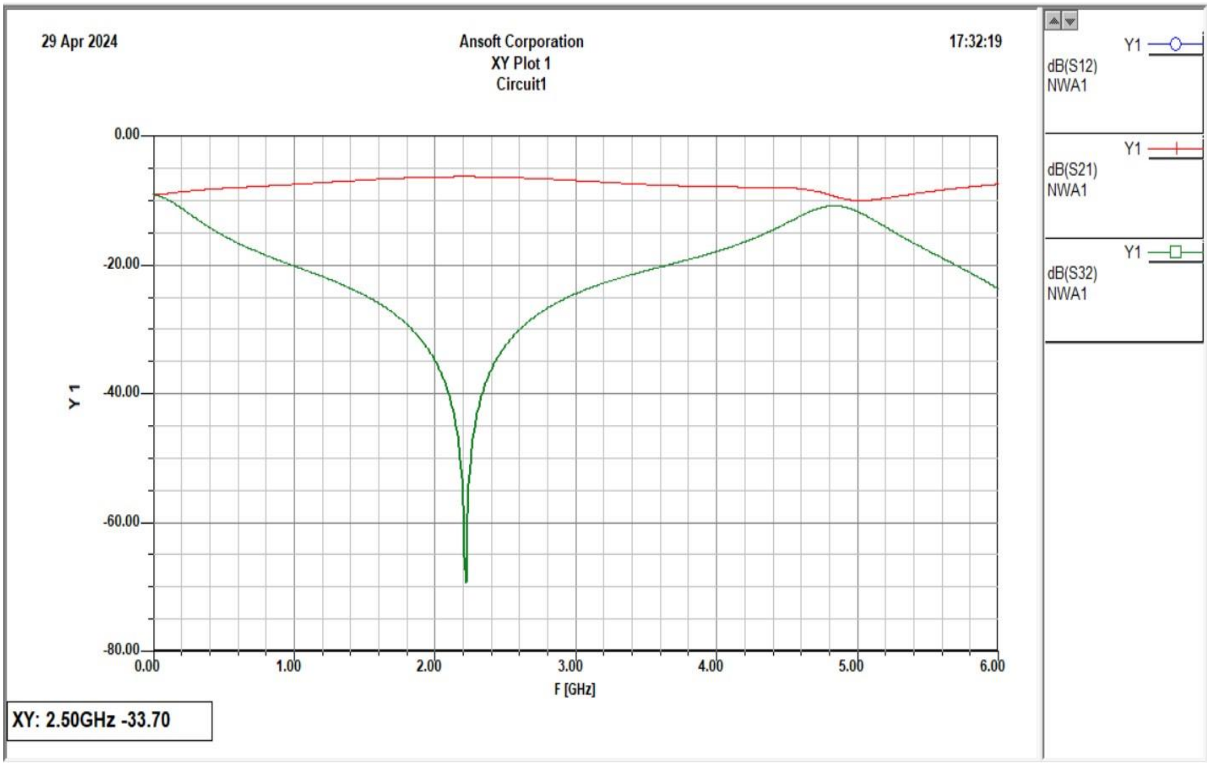
$R_2 = Z_0 k = 50 \times 0.447 \Rightarrow \boxed{22.35\ \Omega}$

$R_3 = \frac{Z_0}{k} \Rightarrow \frac{50}{0.447} \Rightarrow \boxed{111.85\ \Omega}$

(c) Layout with dimensions



(d) Analysis of the results.



IV. Conclusion

We successfully made a Wilkinson unequal power divider with P3/P2 = 1/5 and cutoff frequency = 2.5GHz.

Question 12:

I. Question

Design a branch line coupler with equal power division ratio at 2 GHz.

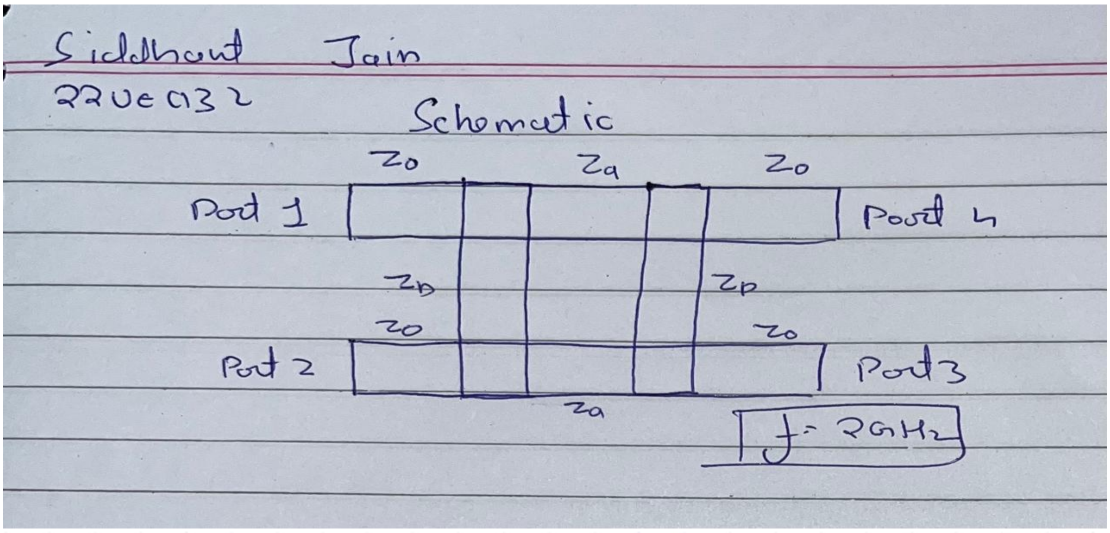
- Show all calculations
- Design the schematic diagram.
- Design the layout on FR4 substrate (4.4) with thickness of 0.8mm.
- Use Ansoft Designer to plot the S_{11}, S_{21}, S_{31}, S_{41}, versus frequency.

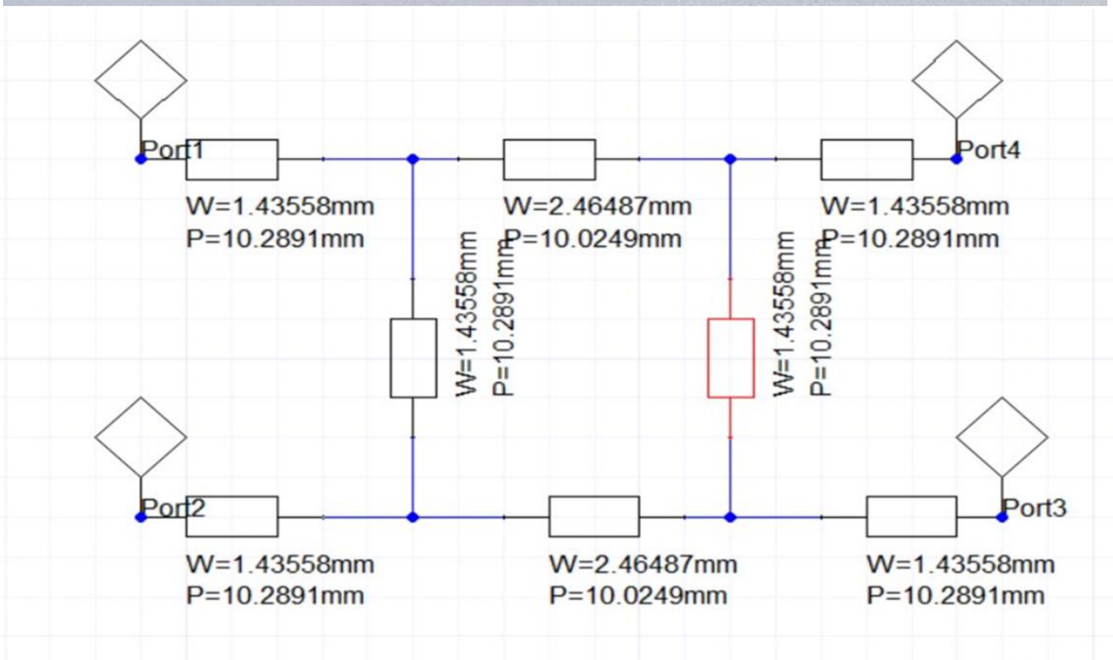
II. Simulation Platform

ANSOFT DESIGNER SOFTWARE

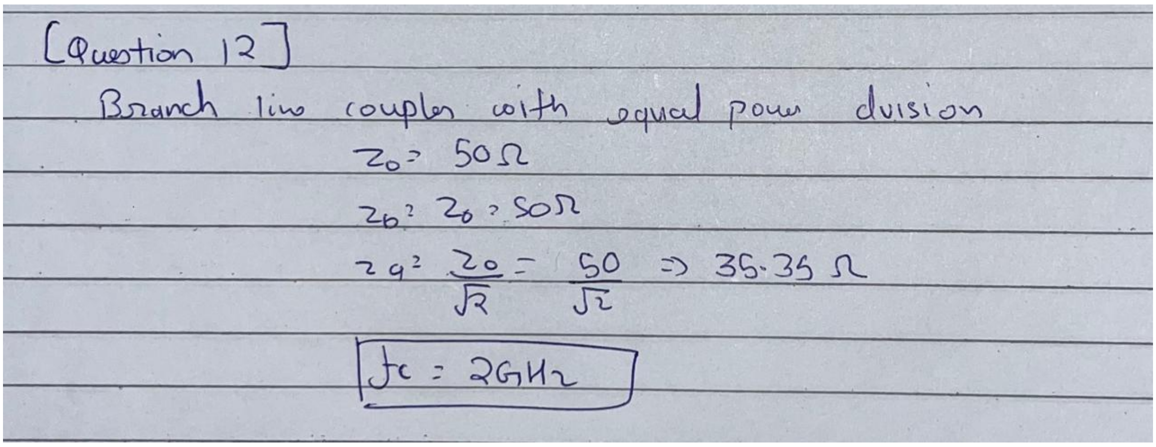
III. Design and Analysis:

a) Screenshots of all schematics





b) Screenshots of Results


[Question 12]

Branch line coupler with equal power division

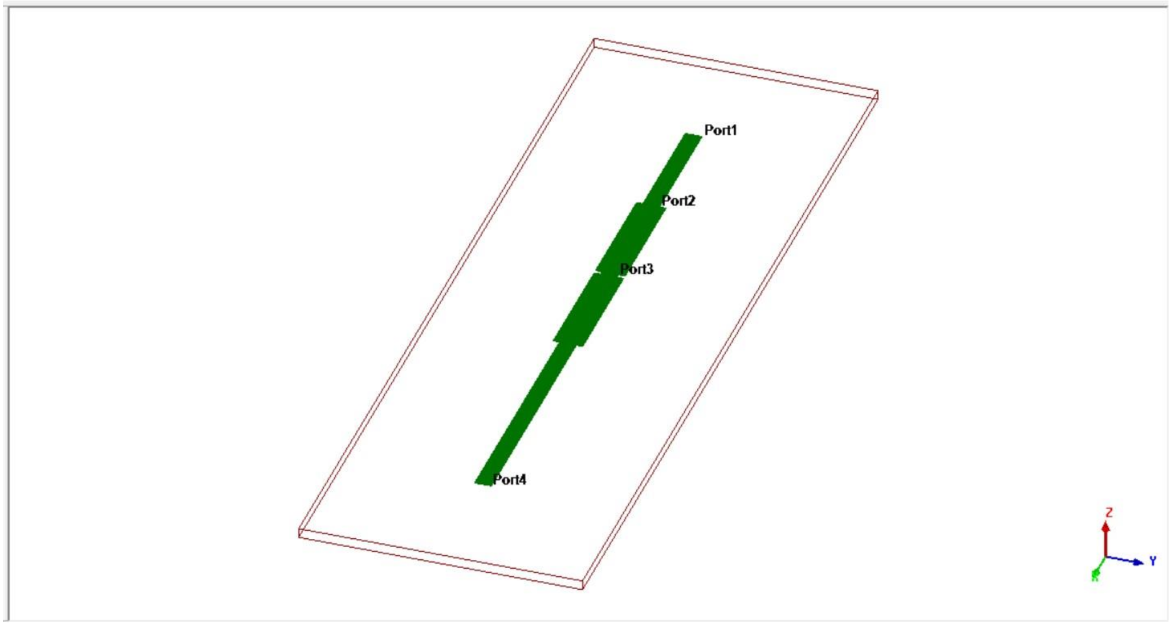
$Z_0 = 50\Omega$

$Z_b = Z_0 = 50\Omega$

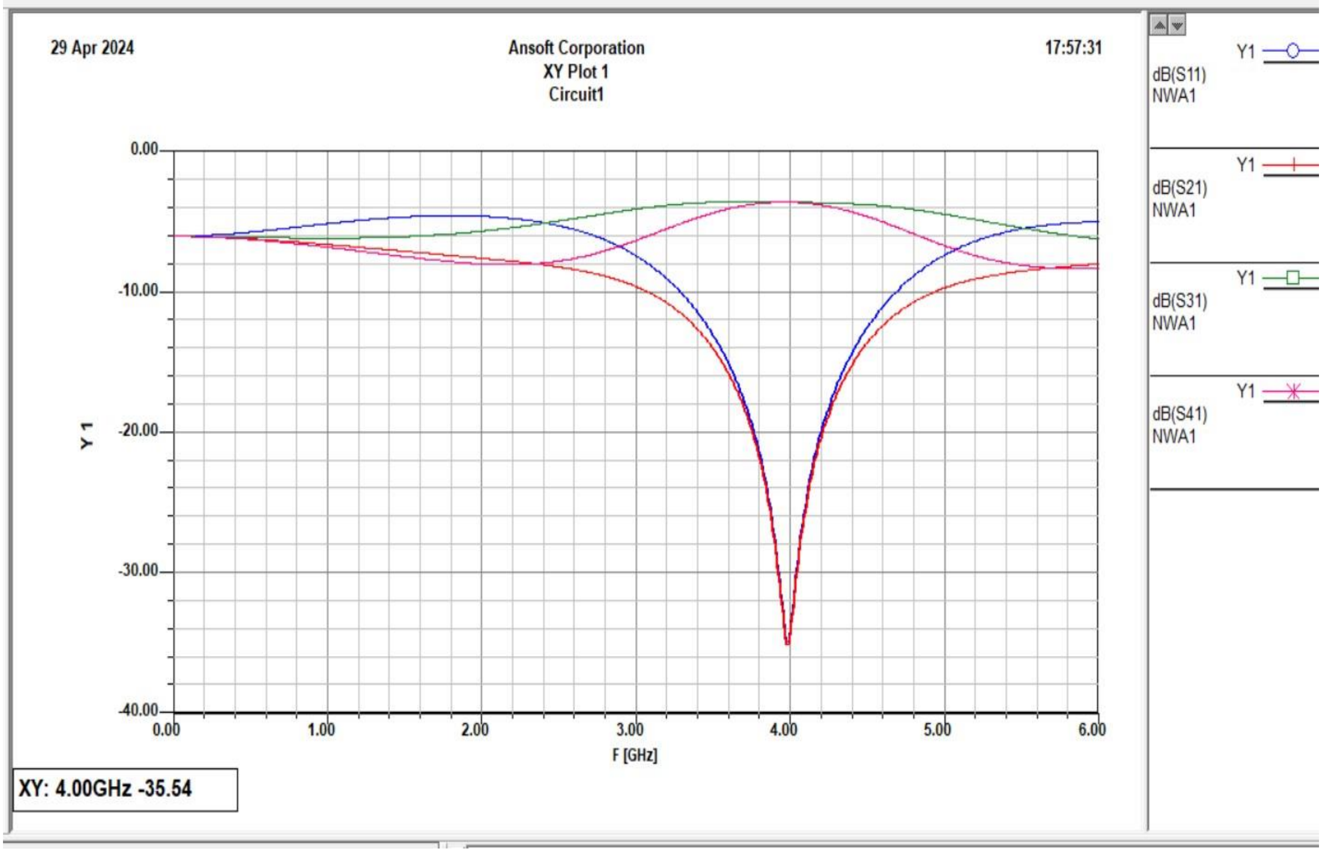
$Z_a = \frac{Z_0}{\sqrt{2}} = \frac{50}{\sqrt{2}} \Rightarrow 35.35\ \Omega$

$\boxed{f_c = 2GHz}$

c) Layout with dimensions



d) Analysis of the results.



IV. Conclusion

We successfully made a Branch line Coupler with Equal power division at 2 GHz with line impedance equal to 50 ohm.